



Prachi Bindal

Roll No. 220102071

B.Tech - Electronics and Communication Engineering

Indian Institute of Technology, Guwahati

+91-9343435162

p.bindal@iitg.ac.in

p.bindal275@gmail.com

GitHub\PrachiBindal | LinkedIn

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	9.02 (Current)	2022-Present
Senior Secondary	MPBSE Board	94.8%	2021
Secondary	MPBSE Board	94.2%	2019

EXPERIENCE

- BNY Mellon** May. 2025 – Jul. 2025
Software Developer Intern / LOR Pune, India
 - Delivered **LLM**-powered system to visualize and analyze **Counterparty Default Loss (CPDL)** financial data from CSV files.
 - Integrated BNY's proprietary agent with **DSPy** to enable **NLP**, **Chain-of-Thoughts**, and prompt tuning via **Python** pipeline.
 - Leveraged **React** for a responsive UI/UX, and **FastAPI** with **CORS** for seamless frontend-backend integration.
 - Received **Pre-Placement Offer**, after successful completion of 8-week onsite internship.

PROJECTS

- $G_m - C$ Filter Tapeout** Jan. 2025 - Apr. 2025
B.Tech Project, Dr. Mahima Arrawatia, IIT Guwahati / LOR bit.ly/GmnBiquad
 - Designed and fabricated a fully differential G_m -cell with **20dB DC gain** and **400MHz bandwidth** for active filter applications.
 - Implemented G_m using **inverter-based Nauta's** architecture in TSMC **180nm** CMOS technology on **Cadence Virtuoso**.
 - Designed a fully differential **biquad** filter to emulate passive **inductors** and resistors with a power consumption of **10 mW**.
 - Realized a 6^{th} order low-pass filter by cascading biquads, resulting in roll-off of **120 dB/dec** with cutoff frequency of **10 MHz**.
 - Utilised **common centroid** matching technique for capacitors layout and performed **post-layout** simulations.
 - Designed an **inverter chain** and **common-drain buffers** to overcome oscilloscope loading, and **decoupling capacitors** were added.
- Bandgap Current Reference Design** Mar. 2025, Bangalore, India
Analog Design Hackathon, CDAC/MeitY bit.ly/CurRefer
 - Designed a bandgap **current reference** circuit in a **90nm** CMOS process, using a **beta-multiplier** core for the bandgap.
 - Achieved temperature coefficient of less than **50 ppm/°C** with current variation of **0.8 μ A** over the range of **−40°C** to **125°C**.
 - Designed for **100 μ A** output current with **load regulation** of **0.2%** for a load voltage range of **0** to **1V** using a **cascode mirror**.
 - Achieved power consumption of **20 μ W** with **1.8V** supply, including **startup circuit**, performed **stb** and **monte carlo** analysis.
 - Performed **layout** using **common-centroid** matching for PNP & current mirror transistors and **serpentine** geometry for resistors.
- Two-Stage Op-Amp Design** Aug. 2024 - Nov. 2024
B.Tech Project, IIT Guwahati bit.ly/FullyOpamp
 - Designed a two-stage **miller compensated** op-amp in TSMC **180nm** technology on **Cadence Virtuoso** for a **1pF** load with **55dB** open-loop **gain**, used a **zero-nulling** resistor for **60° phase margin**, **5 MHz GBW**, **slew rate** of **40 V/ μ s** with **100 μ W power**.
 - Performed **layout** and **post-layout** simulations using **common-centroid** for the input pair and **interdigitated** for current mirror.
 - Designed a **fully differential** operational amplifier in TSMC **65nm** technology with the **single-ended op-amp** for **CMFB**, achieved **50dB** open-loop **gain**, **60° phase margin**, **10MHz** gain bandwidth product (**GBW**), and **250 μ W power** consumption.
- Other Projects**
 - **Analog Calculator** : Developed calculator for arithmetic operations using **op-amps**, **BJTs**, integrated **Arduino** for BODMAS.
 - **Modulation & Estimation** : Modeled digital **modulation** with **pulse shaping** in **GNU Radio**, analyzed spectra, **constellation**, and **eye diagrams**, and estimated **symbol rate**, **CFO**, and modulation using **cyclostationary** and M^{th} power methods.

TECHNICAL SKILLS

- Tools/Frameworks:** Cadence Virtuoso, Synopsys Custom Compiler, LTspice, MATLAB, GNU Radio, DEEDS
- Programming:** C++, C, Python, JavaScript, Arduino IDE

KEY COURSES TAKEN

- Electronics:** Advanced Analog Circuits, Analog Circuits with Lab, Digital Circuits with Lab, Digital Communication, Principle of Communication with Lab, Digital Signal Processing with Lab, Circuit Theory, Control Systems with Lab.
- Mathematics:** Probability and Random Processes, Ordinary & Partial Differential Equations.
- Computer Science:** Artificial Intelligence, Convolutional Neural Networks, Data Structure and Algorithms.

ACHIEVEMENTS

- Analog Design Hackathon:** Secured **3rd** position among **693** teams organized by **CDAC Bangalore & MeitY**. 2025
- Academics:** Secured **AS/AA** grades (10/10) in all Electronics Engineering courses in the **4th** and **5th** semester. 2024
- Micron Mentorship:** Selected among **300 global** participants in the Micron Global Women Mentorship Program. 2024
- Academics:** Secured **AS** grade in Principle of Communication, Digital Signal Processing, and Digital Lab. 2024
- JEE Advanced:** Secured **AIR 5141** among 0.16 million candidates appeared for the test. 2022
- JEE Main:** Secured **AIR 5359** among 1 million candidates. 2022

EXTRA-CURRICULARS

- Social Service:** Joined donation drives, distributing **400+** food packets to Guwahati's underserved communities. 2024
- StartOps, Senior Executive:** Facilitated an MVP workshop for Startup Roar, guiding startup growth. 2023
- Cepstrum Mentorship Program:** Mentored the freshmen students in academics and extra-curricular activities. 2023